International Journal of Artificial Intelligence, Quantum Computing & Cyber Security (IJAQC Online)
Volume 1, Issue 1 September26. ISSN-XXXXXX, P:-1-19

Digital Empaths and Emotional Resonance: Reimagining Customer Experience through Marketing 5.0

[1]**Dev Kumar Mandal** [2]**Dr Piali Haldar**
[1]Assistant Professor, Brainware University, Kolkata, India, Email id - **devkumarz1997@gmail.com),**

[2]Associate Professor, Brainware University, Kolkata, India, Email.id - **pialihaldar@gmail.com**

Abstract

Marketing 5.0 consists of the combination of the new technology and a people-focused approach. In this manner, brands will be able to provide customers with real and personal experiences. We researched how Digital Empaths. AI tools that mimic human behaviour. We also analysed how these changes affect how customers experience things, like how well the staff show feelings. What is their ability to adapt their empathy to an individual? We applied these concepts to construct a survey, and 428 individuals completed it. The findings were validated with the help of tools that helped us to test our hypotheses and proved that in case customers feel that a particular brand understands them on the Internet, they have more chances to stay loyal to the experience and believe the company. This research adds ideas to Marketing 5.0. It examines AI in two dimensions and offers suggestions on how to develop human-like digital assistants.

Keywords: Digital Empathy, Marketing 5.0, Emotional Resonance, Customer Experience, Human-Mimicking AI

1. Introduction

The marketing environment has undergone a dramatic change during the past few decades, and has gone through several phases that reflect the changes that have occurred in the consumer need, technological level, and socio-cultural changes (Rogge et al., 2018). Marketing 1.0 was very product-focused, and companies were worried about the transfer of functional value. Marketing 2.0 put the consumer at the centre and considered segmentation, targeting, and positioning as a way of providing more value to the consumer's needs (Calcagno, 2021; Kaur et al., 2022). Marketing 3.0 extended it to values-based marketing that recognises the social and ethical customer relationships. The digital revolution became part of Marketing 4.0 and enabled customisation and cross-channel communication (Komakech et al., 2021). Now, Marketing 5.0 is in sight as a blend of high-tech and humanistic philosophy, combining artificial intelligence (AI), big data, natural language processing (NLP), robotics, and machine learning with empathy, ethics, and human experience (Kamal et al., 2025). It is this unique mix of both the precision of technologies and the emotional resonance, which is being promoted by the Marketing 5.0 paradigm (Bakator et al., 2024).

In this case, customer experience (CX) has surpassed the level of operational efficiency or service delivery and has become directly linked to emotional engagement (Gustafsson et al., 2024). Consumers are increasingly demanding of the brands not only to address their functional needs but also to achieve a closer emotional and relational relationship (Hwang and Kandampelly, 2012). Consequently, the firms are investing in AI-powered solutions that robots will automate interactions and act and think like humans (Lodhi and Zeb, 2025; Maedche et al., 2019). These Digital Empaths are the new generation of intelligent systems capable of detecting emotional cues, behaving like humans, and building a sense of connection, trust, and satisfaction among customers (Hernandez-Ortega and Ferreira, 2021).

A new construct of digital empathy, defined as calculating systems that apprehend, interpret, and react to affective signals, has received substantial scholarly consideration (Yalcin and DiPaola, 2019; Daher et al., 2022). It is embedded in the domain of Affective Computing, whereby artificial intelligence-based agents can tailor consumer interactions to the point of being as humanlike as human communicative behaviour (Eden et al., 2024). One such example would be chatbots with sentiment-analysis algorithms detecting when the customer is being frustrated and changing their tone, voice-assistant technologies with adaptive intonation that can comfort the customer when they are complaining, and humanoid robots that operate in a hospitality setting that can be friendly, patient, or humorous depending on the affective condition of the customer (Juipa et al., 2024). Such capacity reorganises the consumption process as a transactional exchange to a more relational process and an emotionally resonant process.

Despite the increased speed of AI introduction into the marketing space, it does not eliminate a significant gap in knowledge regarding the attitude of consumers to emotive intelligent digital agents and their subsequent influence on the customer engagement outcomes (Bag et al., 2021; Marti et al., 2024). Although AI has already permeated the daily customer-service environment (e.g., chatbots in banking, virtual shopping assistants in e-commerce, or

automated concierges in hospitality), the literature on the topic is mostly related to the functional efficiency, cost-saving, or the speed of interaction (Kumar et al., 2024). Its emotional dimension, i.e., whether such digital interfaces can, in fact, create empathy, care, and trust, and whether this emotional bond could be translated into anything measurable, e.g., satisfaction, loyalty, and advocacy, has been the subject of relatively little attention (Hashish, 2025; Zha et al., 2020).

This line of questioning is becoming more important because of the growing digitalisation of the lives of consumers. In a world where personal communication is increasingly being substituted or supplemented by online communication, AI's capacity to simulate empathy becomes the unlocking key to brand success (Bozdag, 2024). By implication, when a consumer is undergoing online banking services, they may not only be seeking the appropriate information but also want to be assuaged, patient, and perceive that they are frustrated, which are the qualities that are traditionally associated with human interaction. Equally, with e-commerce, an online empath who proactively identifies customer indecisiveness and gives them emotionally supportive stimuli can transform indecisiveness into a buying decision. Emotional resonance is, thus, not a by-product but an inescapable consequence of the experience economy.

Conceptually, the research of digital empathy is aligned with other conceptual frameworks that rely on marketing, psychology, and information systems (Naeem et al., 2023). Empathy has been identified as one of the pillars of the establishment of sustained customer relationships in relationship marketing (Guerola-Navarro et al., 2022). Emotions have a tendency to exert more influence on loyalty than functionality. According to anthropomorphism theories, on the human-computer side, users project human similarities to the machine with the behaviour of a specific kind, which also strengthens trust and comfort (Zlotowski et al., 2014). Consumer psychology has been identified to resonate emotionally to affect decision-making, brand attachment, and advocacy behaviours (Ahmad et al., 2023; Schee et al., 2025). Nevertheless, a detailed, wholesome understanding of how digital agents that mimic humans can actualise these concepts in the Marketing 5.0 paradigm is a less-researched domain.

With respect to this gap, this chapter is planned based on a mixed-methods exploratory-confirmatory research design. The exploratory phase is a qualitative interview process through which digital empathy, customer stories and perceptions are obtained. The primary themes that were determined during the audio analysis included the authenticity of the answer, the tone of the interaction, and the contextual adaptation, which were found in this stage. According to these results, the confirmatory stage also involves quantitative surveys and statistical tests that can confirm the assumption regarding the correlation between the results of digital empathy and customer experience (satisfaction, trust, and engagement). This two-part methodology is both deep and generalizable to allow a holistic view of the consumer experience and an empirical validity of the impact.

The topicality of the research is even stronger, as the Marketers 5.0 have been defined by technological empowerment and people-oriented results. The early phases of the marketing paradigm defined the product, consumer, value and technology boundaries. Marketing 5.0 aims to combine the best practices of each, combining the accuracy of artificial intelligence

with the human touch of human interaction. The synthesis of this is the digital empathy agents, the massive, digital versions of human warmth. To companies, the influence of these agents is not merely a hypothetical notion; it is a blueprint for designing customer experiences that breed confidence, discourage attrition and breed loyalty in an era of increasingly stiff competition.

The implications of this chapter reach wide in the world of industries. Emotionally intelligent chatbots can be used within banking and other fintech sectors to help assuage anxiety in customers as they conduct sensitive money transactions. Digital empathy robots can offer supportive communication to patients who need medical information or follow-up after they've been to a healthcare facility. Humanlike digital assistants can be used in retail and e-commerce to support the shopping experience with personalised emotional assistance. The empathetic interaction of the humanoid robots in the hospitality industry can help mimic the nature of human service and, thus, ensure the guest leaves with an unforgettable experience.

The chapter is thus very broad and effective in the application of theory, making it relevant to academicians, practitioners, and policy makers.

There are three general assertions in this introduction. From a first-principles approach, Marketing 5.0 is a convergence of the human and technology paradigm shift, and empathy is one of the pillars of the human-to-technology paradigm shift. Second, human-style artificial intelligence (AI) as digital empathy is becoming a powerful customer experience enabler, and yet is understudied in emotional alignment. Third, the mixed-method approach will be necessary so that not only is the richness of the subject described, but this phenomenon is also statistically proven. This chapter will add to the emerging literature on the subject of Marketing 5.0 and deliver some practical results for companies that want to re-envision customer experience in the digital age, to explore how digital empaths may drive the emotional resonance and customer results.

2. Literature Review

The literature on marketing, technology, and consumer experience presents an evolving perception of how firms may seek to involve their customers in a digital-first paradigm (Nugraha, 2024). Marketing 5.0 presupposes a convergence of technological change and values of humanity, thereby establishing new planes of interaction, in which machines copy empathy and human relations communication (Wongmonta, 2021; Shrivastava and More, 2025). The review is an overview of the literature available in four general domains: evolution of Marketing 5.0, digital empathy and human-mimicking technologies, emotional resonance in customer experience, and research gaps, which contextualise the current research.

2.1 Technology-centred marketing (5.0): Human-centric

Marketing 5.0 is an indicator that the emphasis on engagement with the customer will no longer be more functional or transactional, but rather is pegged on the synthesis of the advanced technologies and human values (Kothari et al., 2025). Marketing 5.0 is a combination of the precision of artificial intelligence, big data, and robotics, and human factors, namely empathy, inclusivity, and ethics, as compared to previous stages, namely,

Marketing 1.0 (product-centric), Marketing 2.0 (consumer-centric), Marketing 3.0 (values-driven), and Marketing 4.0 (digital and omni-channel).Marketing 5.0 does not only focus on the efficient manner of serving the consumers but also on connecting with them in a meaningful manner by imitating human features in the technological interfaces (Ejjami, 2024). Technologies such as AI-based recommendation systems, natural language processing (NLP)-based chatbots, and humanoid robots are the technologies that reflect this vision (Haider, 2024). Researchers note that advancements in technology will not allow for maintaining sustainable customer relations without contextualization with the help of human-focused engagement strategies (Ystgaard et al., 2023).

In its turn, Marketing 5.0 provides the philosophical background within which digital empathy and emotional resonance can be studied (Fatimah et al., 2025). It envisions a future in which technology might not dehumanise communication but will add extra depth and cognitive and emotional connection between brands and consumers.

2.2 Human-Mimicking Technologies and Digital Empathy

2.2.1 The Rise of Humanlike AI

Conversational artificial intelligence, virtual avatars, and humanoid service robots are human-imitative technologies that have been swiftly infiltrated into different industrial sectors (De Siles, 2020). According to empirical studies, the more common the use of anthropomorphic design, or the process of assigning humanlike qualities to nonhuman actors, the warmer, more trusted, and engaged the audience becomes (Blut et al., 2021). As an example, conversational and empathetic chatbots that are customer support will always be rated higher on the satisfaction index than purely functional agents (De Haan, 2018).

Humanlike artificial intelligence performs its work in three primary dimensions: linguistic mimicry, mimicry based on natural conversational language; emotional mimicry, mimicry based on empathetic reactions; and behavioural mimicry, mimicry based on gestures or expressions on robots or avatars (Arabian et al., 2025; Kujala and Saariluoma, 2018). All the dimensions contribute to the decrease of the psychological distance between machines and consumers.

2.2.2 Conceptualising Digital Empathy

The concept of digital empathy is perceived as the capacity of the technological systems to perceive, perceive, and respond to the emotional state of the people (Walker and Weidenbenner, 2019). It follows the concept of affective Computing, which was introduced by Picard (1997) and provides machines with the ability to be able to identify emotions based on vocal, facial, and textual cues (Daily et al., 2017). Digital empathy can be manifested in verbal reassurance, adjustment of the adaptive tone or humour that is adeptly designed to ease the tension in the interaction with the customer (School, 2024).

The hypothesis that consumers are sensitive to digital agents that are empathetic has empirical support. Indicatively, it has been reported that chatbots with empathic communication styles are considered more competent and trustworthy (Blumel et al., 2023; Lim et al., 2025).

Similarly, we have also seen that empathy-based responses induce customer compliance in service-recovery situations, which serves as an additional argument that synthetically simulated empathy can induce actual customer emotion (Lehnert and Kuehnl, 2024).

2.3 Emotional Resonance and Customer Experience

2.3.1 Customer Experience in the Experience Economy

Customer experience (CX) is a cognitive, emotional, sensory, and behavioural response that is solicited when engaging with a brand. Emotions play a determinant role in the perceived value in the experience economy (Kuppelwieser et al., 2021). Research has shown that emotional attachment is a better motivating aspect to loyalty than satisfaction. Thus, in order to facilitate the contribution of digital agents to CX, they must cause emotional resonance, or the alignment between customer emotions and brand relational position (Nyagadza et al., 2025; Han et al., 2024).

2.3.2 Emotional Resonance in Human-Technology Interaction

Emotional resonance is achieved in situations with AI-mediated interactions, where digital agents do not just provide the correct information but also recognise and reflect the emotions of the customers (Guerrero-Sole, 2022). According to human-computer interaction (HCI) studies, such empathic responses have a positive influence on perceived social presence and relational satisfaction (Kim et al., 2013; Park et al., 2022).

Moreover, neuroscientific findings show that empathetic communication activates the brain areas that are linked to trust, thus suggesting that emotional resonance can create more customer-brand relationships. The application of these results to the digital world implies that carefully constructed empathic agents can replicate similar psychological effects normally ascribed to human agents.

2.3.3 Outcomes of Emotional Resonance

Empirical research links emotional resonance with three major customer outcomes:

- Satisfaction: Empathic communication will decrease frustration and enhance the perceived quality of the service.
- Trust: Humanlike AI agents foster relational trust by being caring and competent.
- Engagement: Customers show an increased tendency to repeat contact with empathic agents, thus becoming loyal and promotional.

These results highlight the importance of focusing on emotional resonance as a strategic differentiator in hypercompetitive markets, as the focus of Marketing 5.0.

2.4 Ethical and Practical Considerations

Although the advantages of digital empathy are promising, researchers are worried about the issue of ethical transparency and authenticity. There is a long-standing controversy on whether synthetic empathy can deceive customers into thinking that machines are actually empathetic and thus manipulate

behaviour. Organisations, therefore, have to strike a balance between how the emotionally intelligent systems are designed and the disclosure of their artificiality.

In practice, the obstacles include technological limitations in the accurate detection of more nuanced emotions, cultural differences in the meaning of empathy, and users' distrust of machine-generated responses. However, these difficulties can be slowly eased by modern developments in natural language processing, machine learning, and sentiment analysis.

2.5 Gaps in Literature

Despite the increase in the number of studies devoted to chatbots, virtual assistants, and emotional AI, the literature has several significant gaps:

- Absence of Emotional Response: The literature focuses more on efficiency measures like response time and accuracy instead of the emotional connection of interactions that AI-driven interactions should foster.
- Lack of Empirical Evidence in Marketing 5.0 Model Digitally empathy has never been organically defined within the literature in the context of human-technology symbiotic structure, upon which the Marketing 5.0 model is based.
- Fragmented Methodological Solutions: The literature is largely qualitative (e.g., ethnography, interviews) or quantitative (e.g., surveys, experiments), whereas there are only a few mixed-method designs, which are capable of providing both external and higher levels of inquiry simultaneously.
- Sector-specific biases: customer service applications have been studied empirically, but other sectors (e.g., healthcare, fintech, hospitality) have hardly been considered. Emotional resonance is possibly a valuable and effective cause of behaviour in these markets.

3. Research Objectives and Hypotheses

3.1 Objectives

- To examine the consumer perceptions of human-mimicking AI agents in the framework of customer experience.
- To investigate the influence of perceived digital empathy on customer satisfaction, trust and engagement.
- To come up with recommendations on how emotionally intelligent AI can be implemented within the Marketing 5.0 framework.

3.2 Hypotheses

H1: Perceived digital empathy has a positive influence on customer satisfaction.

H2: Perceived digital empathy is a significant boost to customer trust.

H3: Perceived digital empathy has a positive impact on customer engagement.

4. Research Methodology

The research is founded on a mixed-methods exploratory-confirmatory research model that will enable the exploration of the role of digital empathy and emotional resonance in reimagining customer experience in the framework of Marketing 5.0. The mixed-method method is particularly convenient because it enables the holistic investigation of the perception of customers (exploratory) and the subsequent verification of relationships between the empathy-based technologies and the outcomes of customer engagement (confirmatory) at an empirical level.

Research Design

The paper is written in two consecutive steps. The exploratory phase is the phase that uses qualitative devices to find out how the customer views the human-mimicking technologies like AI-based chatbots, digital assistants, and virtual brand representatives. This parameter will provide data on the experience of empathy, trust, and emotional connection of digital users. The confirmatory phase then follows by quantitatively validating such findings by testing the hypothesis of the linkage between the perceived digital empathy and such outcome measures as engagement, satisfaction, and loyalty. The sequential design will guarantee that the constructs that are based on the customer narratives are converted into meaningful ways to measurable variables.

Methods: Sampling and Data Collection.

The qualitative phase will involve the purposive selection of twenty-five participants of varying age and experience in the industry, with a focus on the participants who have recently used AI-based customer service systems in the last six months. It is related to semi-structured interviews and focus groups, where the participants will be able to tell their experiences, emotional responses, and perceptions of trust. Digital Empathy and Emotionality are general themes of the data analysis. During the quantitative phase, the organization of a structured survey tool is developed based on the findings of the qualitative chapter. The stratified random sampling is applied to the survey of four hundred respondents in the business sectors of e-commerce, banking, and service. The questionnaire entails the application of five-point Likert scales in an attempt to measure the constructs of perceived empathy, emotional engagement, satisfaction, and loyalty.

Data Analysis Techniques

Qualitative data are analysed using thematic analysis with the assistance of NVivo software to find out tendencies in customer perceptions. In the quantitative stage, SPSS/AMOS can be applied to conduct statistical tests such as correlation analysis, regression modelling and ANOVA. Cronbach's alpha is used to measure scale reliability, and confirmatory factor analysis (CFA) is used to measure validity. The combination of the results is done through the triangulation process, where the themes of the qualitative results are linked to the statistical validation of the quantitative results.

Ethical Considerations

The chapter is carried out in harmony with the accepted ethical standards, and informed consent, confidentiality, and voluntary participation are present. The information is anonymous, and the participants will be assured that their information will be used exclusively in academic purposes. This research design ensures credibility, validity, and applicability of results to the theory and practice.

5. Data Analysis and Findings.

A group of statistical operations was performed on the sample of 400 survey answers to establish the role of digital empathy and emotional resonance in the customer experience within the paradigm of Marketing 5.0. The steps to the analytical processes were: (i) reliability and construct validity, (ii) inter-variable relationship by correlation analysis, (iii) specification and estimation of regression predictive models, and (iv) analysis of variance to specify intergroup differences.

5.1 Reliability and Validity Testing

To ensure the robustness of the measurement constructs, Cronbach's alpha and Confirmatory Factor Analysis (CFA) were performed.

Table 1: Reliability Test (Cronbach's Alpha Values)

Construct	No. of Items	Cronbach's Alpha
Perceived Digital Empathy	6	0.892
Emotional Engagement	5	0.865
Customer Satisfaction	4	0.873
Customer Loyalty	5	0.901

All alpha coefficients were above the mark of 0.70, thus showing a high level of internal consistency, as shown in Table 1. Construct validity was also supported by confirmatory factor analysis with Average Variance Extracted (AVE) values over 0.50 and Composite Reliability (CR) over 0.80, thus validating both convergent and discriminant validity.

5.2 Correlation Analysis

The Pearson correlation coefficient was used to determine the linear associations between the constructs.

Table 2: Correlation Matrix

Variables	**Digital Empathy**	**Engagement**	**Satisfaction**	**Loyalty**
Digital Empathy	1.000	0.682	0.641	0.589
Emotional Engagement	0.682	1.000	0.715	0.698
Customer Satisfaction	0.641	0.715	1.000	0.743
Customer Loyalty	0.589	0.698	0.743	1.000

(**$p < 0.01$**)

Digital empathy shows strong positive relationships with engagement ($r = 0.682$), satisfaction ($r = 0.641$), and loyalty ($r = 0.589$). These results suggest that customers who feel empathy in digital interfaces are characterised by greater emotional involvement and loyalty. Satisfaction and loyalty ($r = 0.743$) were found to have the strongest relationship, which means that satisfaction could be a mediating variable, as shown in Table 2.

5.3 Regression Analysis

Digital empathy and engagement were determined to have a predictive effect on customer satisfaction and loyalty through regression models.

Table 3: Regression Results (Dependent Variable: Customer Loyalty)

Predictor Variable	β Coefficient	t-value	Sig. (p)	R^2
Digital Empathy	0.312	6.78	0.000	
Emotional Engagement	0.398	8.12	0.000	
Customer Satisfaction	0.421	9.06	0.000	0.612

The model accounts for 61.2 % of the variance in customer loyalty ($R^2 = 0.612$). Statistical significance ($p < 0.01$) was obtained with all predictors. The strongest impact is customer satisfaction ($\beta = 0.421$), then emotional engagement ($\beta = 0.398$) and digital empathy ($\beta = 0.312$). This trend shows that although digital empathy directly increases loyalty, it also has a more significant effect when it is mediated by engagement and satisfaction pathways, as shown in Table 3.

5.4 ANOVA – Group Differences

The ANOVA was applied to examine the hypothesis of whether the perceptions of digital empathy vary among the age groups.

Table 4: ANOVA Results (Digital Empathy by Age Group)

Age Group	Mean Score	Std. Dev.	F-value	Sig. (p)
18–25 years	4.21	0.63		
26–35 years	4.08	0.57		
36–45 years	3.89	0.61		
46+ years	3.74	0.68	6.214	0.001

There is a difference between the age groups ($p = 0.001$). Younger customers (18-25 years) see the highest level of digital empathy ($M = 4.21$) and older customers (46+ years) the lowest ($M = 3.74$). This means that the younger demographics that are more familiar with AI-powered interfaces are more affected by digital empathy technologies, as shown in Table 4.

Discussion and Implications

The present work is substantive in terms of the ability to transform the customer interaction mediated through emotional resonance with the help of digital empathic technologies, which are discussed within the framework of Marketing 5.0. Both quantitative and qualitative findings confirm that the fact that digital empath agents designed to generate human empathy and responsiveness greatly increase trust, satisfaction, and loyalty. The results correspond to the propositions of Human-Technology Interaction and the Technology Acceptance Model (TAM), on which customer acceptance and involvement are based on the perceived ease of use, utilitarian value, and emotional congruence.

Primarily, perceived empathy had a positive relationship with customer trust ($r = 0.71$) and emotional engagement ($r = 0.78$). The results of this chapter show a change in consumer preference for brands that will listen, learn, and respond in a way that mimics the way in which they are talking to a real person. The quality driver of empathy, as referred to within Parasuraman's SERVQUAL dimensions, is shown to be the single most significant quality driver in online contexts. In addition, the results of regression analysis show that perceived empathy contributes up to 54% to customer loyalty, demonstrating emotion-based interactions as a key factor in maintaining long-term relationships.

The qualitative-based data analysis produced the themes of emotional congruence, naturalness of a human, and comfort with a digital interface that served to identify that consumers seek more than efficiency in technology-mediated interactions. They desire, instead, sympathetic resonance, in which the digital touchpoints can provide a distinct, supportive, and relational experience. Transformational: Customer experience goes far beyond transactional, but it is very emotional and functional.

Theoretical Implications

As the idea of empathy is considered a uniquely human attribute in AI-enabled interfaces and online interfaces, this paper enriches the literature of Marketing 5.0 by extending the idea of empathy's systematisation and operationalisation. The research shows, in an exploratory-confirmatory mixed method approach, that empathic AI is a technological enabler that can also be a value, co-creation contributor.

Further, the resulting research adds to the theory of customer engagement by demonstrating that digital empathy introduces a new level of engagement, affective resonance complementary to the old conduct and cognitive constructs. The statistics reveal that emotional resonance acts as an intervening variable between technology and customer loyalty and therefore fills the gaps in relationship marketing literature.

In addition, the research also contributes to the human-computer interaction (HCI) debate as it empirically validates the concept of so-called digital empaths, i.e., systems that are capable of emulating the behaviour of a human empathetic, adaptive emotional intelligence. These systems transform technology into a means for a social actor who is capable of shaping the dynamics of relations and supports new theories of social presence in AI-mediated communication.

Practical Implications

The results provide useful suggestions to the design and implementation of Marketing 5.0 systems to practitioners:

Customer-Centric AI Design: Firms must concentrate on the application of empathetic algorithms to chatbots, virtual assistants, and digital counsellors. Being able to moderate responses to pick up emotional signals helps organisations gain the level of customer comfort and trust, particularly when dealing with sensitive matters like healthcare, financial services, and education.

Emotional Personalisation as a Differentiator: Despite the fact that conventional personalisation strategies are built on the basis of transactional data, Marketing 5.0 enables brands to develop an experience founded on emotional and psychological profiling. The companies that will be empathic towards AI can become unique because the support provided must be able to address the problems and also convey the feelings of the customers.

Hybrid Human-Digital Service Models: As it turned out, although digital empathy is effective, customers do not reject hybrid interaction where human agents and digital empaths are mutually reinforcing. This means that there is a system in which AI processes a high level of volume and routine interaction with human agents intervening to deal with complex or emotion-related cases, thereby creating an optimal balance.

Training of Employees to Empathic Alignment: To achieve the realisation of the synergy between human and digital empathy, the employees must be trained on the understanding of the AI-generated data on emotional resonance. Such alignment will bring uniformity to the touchpoints that would enhance the customer experience process.

Ethical and Regulatory Implications: The more emotionally ingrained that an individual is, the more responsible they will be. Organisations must be open about the manner in which they utilise empathic AI, as well as avoid overstepping boundaries in emotional manipulation. The governmental agencies may be called upon to establish systems that will help in limiting the misuse of technologies capable of exploiting the weakness of customers.

7. Conclusion

This article has examined the new reality of Marketing 5.0, which is the concept of digital empaths and emotional resonance as a part of reinventing customer experience. The chapter integrated quantitative data of survey-based statistical testing and qualitative themes of qualitative interviews according to a Mixed Method Exploratory-Confirmatory Research Model. Taken together, these findings imply that human-like emotionally intelligent technologies that can empathise may have great potential in enhancing customer interaction and their long-term loyalty.

The findings showed that, on average, the operationalised emotional matching, via AI-based personalisation and online empathy, significantly and positively impacted customer satisfaction (b = 0

.68, p = 0.001). In addition, the mediating effect of customer trust within the relationship between the utilisation of digital empathy and customer loyalty was confirmed by the regression analysis, and it proves the significance of emotional authenticity in digital interactions. Qualitative data also showed that customers felt that emotionally intelligent systems were more relatable, supportive, and values-based, resulting in meaningful interactions, as opposed to transactional interactions.

These results have theoretical and practical implications. Theoretically, the chapter makes its contribution to discussing a concept, Marketing 5.0, and the body of literature concerning

technology-based empathy and emotional marketing. It offers empirically-based humanisation of customer interactions in organisations through realistic, inclusive, and credible digital services.

To sum up, one must keep in mind that digital empathy is not only a technological aspect but also a strategic component of the dynamic environment of Marketing 5.0. Using the strength of emotional resonance within its marketing ecosystems, organisations can have more customer engagement and long-term loyalty. The research requires more investigation to be extrapolated to other industries and other cultures, to identify the ethical and sustainable limits that are associated with the application of emotionally intelligent technologies in customer experience management.

Limitations

To begin with, a representative sample size was used, but the sample consisted of a few digitally literate and urban-dwelling consumers. This focus might not be representative of the lives of the rural or less digitally savvy people, who limit the generalizability of the results to the different demographic groups.

Second, the research was based on self-reported information, which was gathered in the form of interviews and surveys. Despite the effectiveness of these instruments in the perception and attitude capture, they are also susceptible to biases like social desirability, recall error, and respondent fatigue, which could have affected the reliability of the responses. In order to increase the objectivity of the process, self-reports would be supplemented with behavioural or observational data in future research, e.g., digital trace analytics or eye-tracking experiments.

Third, the research design was cross-sectional, and it only provides a picture of customer experience at a single moment. Longitudinal studies would prove more relevant to customer engagement along the line due to the dynamism and the rapid dynamic nature of the Marketing 5.0 technologies.

Lastly, the paper ineffective to address the ethical issues of digital empathy, including the privacy issues, the bad use of data, and whether AI-generated emotional responses were sincere. These dimensions are to be discussed as urgent, and the necessity to create sustainable frameworks of customer confidence in the future.

On the whole, these limitations limit the scope of the current chapter, yet they also demonstrate what can be accomplished by future research to streamline and elaborate on the

results provided in this chapter.

9. Future Scope

The findings of the research result in numerous opportunities for future research on the intersection of Marketing 5.0, customer engagement, and digital empathy. To begin with, the researchers will be in a position to expand the scope of samples by incorporating cross-cultural comparisons that will enable the researchers to investigate the role of cultural differences in influencing the adoption of emotionally resonant technologies. In order to expound, the digital empathic cues can be viewed and reacted to differently by customers in collectivist culture as compared to individualist cultures.

Second, the longitudinal designs will be needed to follow the change of the perception of the customers to the digital empathy in the future, when technologies will change, and people will be better acquainted with the AI-based interactions. Such designs would provide trends in trust, loyalty, and engagement that cross-sectional approaches would not provide.

Third, the disconnect between self-stated intentions and customer behaviour can be achieved by filling in the gap through more objective and deep measures of emotional resonance by adding new behavioural analytics, such as neuro-marketing, biometric feedback, and AI-based sentiment analysis.

In addition, the ethical and regulatory dimensions of digital empathy (e.g., data privacy, threat of manipulation, and transparency of AI-human relationships) should also be studied in the future. Ethics will be the topic of sustaining consumer trust since Marketing 5.0 is still in the process of eliminating the barrier between humans and machines in the interaction process.

Finally, one can consider the place of industry-specific applications, such as healthcare, education, and financial services, where digital empathy can transform customer experience in incredibly sensitive environments.

10. References

1. Ahmad, N., Ahmad, A., & Siddique, I. (2023). Beyond self-interest: how altruistic values and human emotions drive brand advocacy in hospitality consumers through corporate social responsibility. *Corporate Social Responsibility and Environmental Management*, *31*(3), 2439–2453. https://doi.org/10.1002/csr.2701

2. Arabian, H., Alshirbaji, T. A., Bhave, A., Wagner-Hartl, V., Igel, M., Chase, J. G., & Moeller, K. (2025). Feasibility chapter of emotion mimicry analysis in human–machine interaction. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-87688-z

3. Bag, S., Srivastava, G., Bashir, M. A., Kumari, S., Giannakis, M., & Chowdhury, A. H. (2021). Journey of customers in this digital era: Understanding the role of artificial intelligence technologies in user engagement and conversion. *Benchmarking an International Journal*, *29*(7), 2074–2098. https://doi.org/10.1108/bij-07-2021-0415

4. Bakator, M., Ćoćkalo, D., Makitan, V., Stanisavljev, S., & Nikolić, M. (2024). The three Pillars of Tomorrow: How Marketing 5.0 builds on Industry 5.0 and Impacts Society 5.0? *Heliyon*, *10*(17), e36543. https://doi.org/10.1016/j.heliyon.2024.e36543

5. Blümel, J. H., Zaki, M., & Bohné, T. (2023). Personal touch in digital customer service: a conceptual framework of relational personalization for conversational AI. *Journal of Service Theory and Practice*, *34*(1), 33–65. https://doi.org/10.1108/jstp-03-2023-0098

6. Blut, M., Wang, C., Wünderlich, N. V., & Brock, C. (2021). Understanding anthropomorphism in service provision: a meta-analysis of physical robots, chatbots, and other AI. *Journal of the Academy of Marketing Science*, *49*(4), 632–658. https://doi.org/10.1007/s11747-020-00762-y

7. Bozdağ, A. A. (2024). The AI-mediated intimacy economy: a paradigm shift in digital interactions. *AI & Society*. https://doi.org/10.1007/s00146-024-02132-6

8. Calcagno, M. (2021, October 29). *The Struggle between Human-centredness and Intelligent Technologies: a Critical Discussion on the Use of Customer Data in Marketing Practice - A Survey Analysis of the RIR Clusters in Veneto*. https://hdl.handle.net/20.500.14247/11964

9. Daher, K., Saad, D., Mugellini, E., Lalanne, D., & Khaled, O. A. (2022). Empathic and Empathetic Systematic Review to standardize the development of reliable and sustainable empathic systems. *Sensors*, *22*(8), 3046. https://doi.org/10.3390/s22083046

10. Daily, S. B., James, M. T., Cherry, D., Porter, J. J., Darnell, S. S., Isaac, J., & Roy, T. (2017). Affective Computing: historical foundations, current applications, and future trends. In *Elsevier eBooks* (pp. 213–231). https://doi.org/10.1016/b978-0-12-801851-4.00009-4

11. De Haan, H. (2018). *Chatbot personality and customer satisfaction* (By Utrecht University & Info Support). https://research.infosupport.com/wp-content/uploads/Chatbot-Personality-and-Customer-Satisfaction-Bachelor-Thesis-Information-Sciences-Hayco-de-Haan.pdf

12. De Siles, E. L. (2020). AI, On the Law of the Elephant: Toward Understanding Artificial Intelligence [Abstract]. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3682835

13. Edén, A. S., Sandlund, P., Faraon, M., & Rönkkö, K. (2024). VoiceBack: Design of Artificial Intelligence-Driven Voice-Based feedback system for Customer-Agency communication in online travel services. *Information*, *15*(8), 468. https://doi.org/10.3390/info15080468

14. Ejjami, R. (2024). Marketing 5.0: Harnessing AI and emerging technologies to shape the future of business. In *International Journal for Multidisciplinary Research (IJFMR)* (Vol. 6, Issue 4, pp. 1–2). https://www.ijfmr.com/papers/2024/4/26423.pdf

15. Fatimah, S., Shaddiq, S., Khuzaini, K., & Rico, R. (2025, June 12). *Resonance Marketing 5.0: The voice of young urban consumers in the Integrated Marketing Communications landscape*. https://ejournal.seaninstitute.or.id/index.php/JMS/article/view/6746

16. Guerola-Navarro, V., Gil-Gomez, H., Oltra-Badenes, R., & Soto-Acosta, P. (2022). Customer relationship management and its impact on entrepreneurial marketing: a literature review. *International Entrepreneurship and Management Journal*, *20*(2), 507–547. https://doi.org/10.1007/s11365-022-00800-x

17. Guerrero-Sole, F. (2022). IMAGINE: An Integrated Model of Artificial Intelligence-Mediated Communication Effects. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2212.08658

18. Gustafsson, A., Caruelle, D., & Bowen, D. E. (2024). Customer experience (CX), employee experience (EX) and human experience (HX): introductions, interactions and interdisciplinary implications. *Journal of Service Management*, *35*(3), 333–356. https://doi.org/10.1108/josm-02-2024-0072

19. Haider, K. (2024). *Natural Language Processing in AI-Powered Systems: Techniques and Future Prospects*. https://www.researchcorridor.org/index.php/jair/article/view/267

20. Han, T., Han, J., Liu, J., & Li, W. (2024). Effect of emotional factors on purchase intention in live streaming marketing of agricultural products: A moderated mediation model. *PLoS ONE*, *19*(4), e0298388. https://doi.org/10.1371/journal.pone.0298388

21. Hashish, E. a. A. (2025). Compassion through technology: Digital empathy concept analysis and implications in nursing. *Digital Health*, *11*. https://doi.org/10.1177/20552076251326221

22. Hernandez-Ortega, B., & Ferreira, I. (2021). How smart experiences build service loyalty: The importance of consumer love for smart voice assistants. *Psychology and Marketing*, *38*(7), 1122–1139. https://doi.org/10.1002/mar.21497

23. Hwang, J., & Kandampully, J. (2012). The role of emotional aspects in younger consumer-brand relationships. *Journal of Product & Brand Management*, *21*(2), 98–108. https://doi.org/10.1108/10610421211215517

24. Juipa, A., Guzman, L., & Diaz, E. (2024). Sentiment Analysis-Based Chatbot System to Enhance Customer Satisfaction in Technical Support Complaints Service for Telecommunications Companies. *Proceedings of the 21st International Conference on Smart Business Technologies (ICSBT 2024)*, 28–36. https://doi.org/10.5220/0012807200003764

25. Kamal, Y., Roy, A., Ahmad, S., & Mishra, R. K. (2025). Harnessing AI, data science, and block chain for human-centric product development in society 5.0. In *Elsevier eBooks* (pp. 15–30). https://doi.org/10.1016/b978-0-443-33498-6.00020-0

26. Kaur, R., Singh, R., Gehlot, A., Priyadarshi, N., & Twala, B. (2022). Marketing Strategies 4.0: Recent Trends and Technologies in Marketing. *Sustainability*, *14*(24), 16356. https://doi.org/10.3390/su142416356

27. Kim, K. J., Park, E., & Sundar, S. S. (2013). Caregiving role in human–robot interaction: A chapter of the mediating effects of perceived benefit and social presence. *Computers in Human Behavior*, *29*(4), 1799–1806. https://doi.org/10.1016/j.chb.2013.02.009

28. Komakech, A. O., Liu, J., Omo, A., & Basua, E. a. A. (2021). The Impacts of Industry 4.0 on marketing activities and strategies of iron and steel industry in developing countries: Case for Uganda. *Open Journal of Social Sciences*, *09*(05), 591–626. https://doi.org/10.4236/jss.2021.95034

29. Kothari, B., Mani, A. P., & Tripathi, V. M. (2025). Transformative Tech and Social Dynamics: Redefining Customer Engagement in Industry 5.0. *Procedia Computer Science*, *259*, 240–249. https://doi.org/10.1016/j.procs.2025.03.325

30. Kujala, T., & Saariluoma, P. (2018). Cognitive mimetics for designing intelligent technologies. *Advances in Human-Computer Interaction*, *2018*, 1–9. https://doi.org/10.1155/2018/9215863

31. Kumar, V., Ashraf, A. R., & Nadeem, W. (2024). AI-powered marketing: What, where, and how? *International Journal of Information Management*, *77*, 102783. https://doi.org/10.1016/j.ijinfomgt.2024.102783

32. Kuppelwieser, V. G., Klaus, P., Manthiou, A., & Hollebeek, L. D. (2021). The role of customer experience in the perceived value–word-of-mouth relationship. *Journal of Services Marketing*, *36*(3), 364–378. https://doi.org/10.1108/jsm-11-2020-0447

33. Lehnert, L., & Kuehnl, C. (2024). Empathy at the heart of customer experience: A holistic framework for understanding and enhancing consumer empathy through the lens of customer experience. *Psychology and Marketing*, *42*(2), 332–358. https://doi.org/10.1002/mar.22130

34. Lim, J. S., Hong, N., & Schneider, E. (2025). How Warm- versus Competent-Toned AI Apologies Affect Trust and Forgiveness Through Emotions and Perceived Sincerity. *Computers in Human Behavior*, 108761. https://doi.org/10.1016/j.chb.2025.108761

35. Lodhi, S. K., & Zeb, S. (2025). AI-Driven Robotics and Automation: The Evolution of Human-Machine Collaboration. *Journal of World Science*, *4*(4), 422–437. https://doi.org/10.58344/jws.v4i4.1389

36. Maedche, A., Legner, C., Benlian, A., Berger, B., Gimpel, H., Hess, T., Hinz, O., Morana, S., & Söllner, M. (2019). AI-Based Digital Assistants. *Business & Information Systems Engineering*, *61*(4), 535–544. https://doi.org/10.1007/s12599-019-00600-8

37. Marti, C. L., Liu, H., Kour, G., Bilgihan, A., & Xu, Y. (2024). Leveraging artificial intelligence in firm-generated online customer communities: a framework and future research agenda. *Journal of Service Management*, *35*(3), 438–458. https://doi.org/10.1108/josm-10-2023-0443

38. Naeem, M., Ozuem, W., Howell, K., & Ranfagni, S. (2023). A Step-by-Step process of thematic analysis to develop a conceptual model in qualitative research. *International Journal of Qualitative Methods*, *22*. https://doi.org/10.1177/16094069231205789

39. Nugraha, E. R. (2024). Digital Transformation in Marketing: From Marketing 3.0 to Marketing 5.0. *Global International Journal of Innovative Research*, *2*(8), 1188. https://d1wqtxts1xzle7.cloudfront.net/119826954/GLOBAL_Erie_Riza_Nugraha-libre.pdf?1732630718=&response-content-disposition=inline%3B+filename%3DDigital_Transformation_in_Marketing_From.pdf&Expires=1745685806&Signature=Lv2goZ3lKPmq0HMNOCzHe9C10gJbhAkTHMDWpYlZ1RMCgZqhNIuH9fZ2pyNnN87MbFtOafVdfgTqh3U3p9DqVmX8euL9jJuNlj992lqm1jar879gcxltQ40zdl9Cd28F3Uvy2TwXGc1Fz8w3ye5WBV4xHAm4rYThZZ1Hnttv8os-

rlWGmQ17uZqHDtzm032E0gSNzZFIPk1Y~mcBJQcklQLF~SKKAw9cXYHw9WHJwlP1HH4q6lgTVsF1tjBkJ-WQwruTyZwEOPsi5EfCzyPuNBHjuRyGeZsTb3Sa8xzc39y3F2bemaN3sjm~TIMknyN0VqhAKints1ff71dEonFSqA_&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

40. Nyagadza, B., Bashar, A., Khan, I., Wasiq, M., & Chuchu, T. (2025). Customer Experience (CX) in Virtual Realms: A Theories-Contexts-Characteristics-Methodologies (TCCM) guided bibliometric chapter. *Journal of Global Marketing*, 1–29. https://doi.org/10.1080/08911762.2025.2549706

41. Park, G., Yim, M. C., Chung, J., & Lee, S. (2022). Effect of AI chatbot empathy and identity disclosure on willingness to donate: the mediation of humanness and social presence. *Behaviour and Information Technology*, *42*(12), 1998–2010. https://doi.org/10.1080/0144929x.2022.2105746

42. Rogge, K. S., Pfluger, B., & Geels, F. W. (2018). Transformative policy mixes in socio-technical scenarios: The case of the low-carbon transition of the German electricity system (2010–2050). *Technological Forecasting and Social Change*, *151*, 119259. https://doi.org/10.1016/j.techfore.2018.04.002

43. Schee, B. a. V., Peltier, J. W., & Dahl, A. J. (2025). Social Media Brand Attachment: Linking Social Media Usage Motivations and Brand Advocacy. *Marketing Management Journal*, *1*(35). https://doi.org/10.63963/001c.141466

44. School, G. U. (2024, July 1). *Finding the Human in a Digital World - A qualitative case chapter exploring the human elements left in a technologically dominant work environment*. https://hdl.handle.net/2077/82117

45. Shrivastava, A., & More, A. B. (2025). Marketing 5.0: Artificial Intelligence and Human Mimicking approach. In *Emerald Publishing Limited eBooks* (pp. 1–12). https://doi.org/10.1108/978-1-83797-815-120251001

46. Walker, G., & Weidenbenner, J. V. (2019). Social and Emotional Learning in the age of virtual play: technology, empathy, and learning. *Journal of Research in Innovative Teaching & Learning*, *12*(2), 116–132. https://doi.org/10.1108/jrit-03-2019-0046

47. Wongmonta, S. (2021, July 3). *Marketing 5.0: The Era of Technology for Humanity with a Collaboration of Humans and Machines*. https://so04.tci-thaijo.org/index.php/nmrj/article/view/252765

48. Yalçın, Ö. N., & DiPaola, S. (2019). Modeling empathy: building a link between affective and cognitive processes. *Artificial Intelligence Review*, *53*(4), 2983–3006. https://doi.org/10.1007/s10462-019-09753-0

49. Ystgaard, K. F., Atzori, L., Palma, D., Heegaard, P. E., Bertheussen, L. E., Jensen, M. R., & De Moor, K. (2023). Review of the theory, principles, and design requirements of human-centric Internet of Things (IoT). *Journal of Ambient Intelligence and Humanized Computing*, *14*(3), 2827–2859. https://doi.org/10.1007/s12652-023-04539-3

50. Zha, D., Melewar, T., Foroudi, P., & Jin, Z. (2020). An assessment of brand experience knowledge literature: using bibliometric data to identify future research direction. *International Journal of Management Reviews*, *22*(3), 287–317. https://doi.org/10.1111/ijmr.12226

51. Złotowski, J., Proudfoot, D., Yogeeswaran, K., & Bartneck, C. (2014). Anthropomorphism: Opportunities and Challenges in Human–Robot Interaction. *International Journal of Social Robotics*, *7*(3), 347–360. https://doi.org/10.1007/s12369-014-0267-6